Kuta Software - Infinite Calculus

Name____________________

Substitution for Definite Integrals

Date________________ Period____

Express each definite integral in terms of *u*, but do not evaluate.

1) $\int_{-1}^{0} \frac{8x}{(4x^2+1)^2}\, dx;\ u = 4x^2 + 1$

2) $\int_{0}^{1} -12x^2(4x^3-1)^3\, dx;\ u = 4x^3 - 1$

3) $\int_{-1}^{2} 6x(x^2-1)^2\, dx;\ u = x^2 - 1$

4) $\int_{0}^{1} \frac{24x}{(4x^2+4)^2}\, dx;\ u = 4x^2 + 4$

Evaluate each definite integral.

5) $\int_{-3}^{0} -\frac{8x}{(2x^2+3)^2}\, dx;\ u = 2x^2 + 3$

6) $\int_{0}^{1} \frac{16x}{(4x^2+4)^2}\, dx;\ u = 4x^2 + 4$

7) $\int_{-1}^{0} 18x^2(3x^3+3)^2\, dx;\ u = 3x^3 + 3$

8) $\int_{0}^{1} -\frac{8x}{(4x^2+2)^2}\, dx;\ u = 4x^2 + 2$